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"silence" in Tamil, Kannada and Hindi

mouna

Mouth it. Mouna says it.

A phone app that lets a person who cannot make sound speak by mouthing words, in the caregiver's language, offline.

89.0% top-1 on 7 Kannada speakers · whole encoder 18.4 ms on the iQOO 15's NPU · no internet permission

Team Highest in the Room · Maadhav V H · Sachin A S · Nakul T

Open Innovation track · built phone-first for the iQOO 15 (Snapdragon NPU)

[Try the prototype: mouna-spike.vercel.app](https://mouna-spike.vercel.app)



SUMMARY

Mouna lets people who have lost their voice speak by mouthing, offline, in Indian languages

PROBLEM	After laryngectomy, tracheostomy or stroke, many patients can move their lips but make no sound.	~70,000 new laryngeal and hypopharyngeal cancers a year in India (GLOBOCAN 2024)
PRODUCT	Teach three phrases in about 25 s by mouthing them; the phone then speaks each one in the caregiver’s language, in a natural voice the patient chooses.	Kannada, Tamil, Hindi, English; mouth in one, be heard in another
PROOF	With five samples per phrase it reads 89.0% of words and phrases right first time; 97.3% of two-word phrases with three.	7 Kannada speakers, 27 classes, public CC BY dataset; right 94.3% of the times it chose to speak
PLATFORM	The whole lip encoder runs on the Hexagon NPU of the iQOO 15’s chip: 18.4 ms for 1.92 s of lips, every layer on the NPU.	Galaxy S26 (same SM8850) via Qualcomm AI Hub; int8 model 86.3 MB; no internet permission
RESPONSIBILITY	It asks instead of guessing, confirms urgent phrases, stores no video, credits LipLearner and states its limits.	Communication aid, not a medical device; pilot with a speech-language pathology team next

After throat surgery or a stroke, many patients can still move their lips but make no sound

TODAY

- A ward at 3 a.m. The patient needs water, or is in pain.
- Hands are weak or taped to a line; the whiteboard is slow and the script may not be theirs.
- The nurse may speak another language than the patient.
- Voice prostheses help many people later, but there is a gap before they work.

WITH MOUNA

The patient mouths "I need water" in Tamil.

எனக்கு தண்ணீர் வேண்டும்



The phone says it aloud, in Kannada, for the nurse:

ನನಗೆ ನೀರು ಬೇಕು

The need is large, and the one clinical lip-reading app struggles exactly where it is needed

~70,000

new laryngeal and hypopharyngeal cancers a year in India

GLOBOCAN 2024, India fact sheet

51.2%

of stroke patients show speech disturbance at onset (34,792 cases, 30 centres)

Mathur et al., Indian hospital stroke registries

21.8%

accuracy of the cloud lip-reading app SRAVI in UK critical care; Wi-Fi limits hurt usability

HRA summary of the SRAVI study

What the evidence says: patients after laryngectomy preferred lip-reading to writing (69.7%, n = 16), but a fixed, cloud, English-only, speaker-independent model is too weak in a ward. Personalised, on-device, multilingual is the gap.

Teach three phrases in about 25 seconds, mouth them silently, and the caregiver hears them in their own language

1

Teach

Mouth each phrase three times. A quality gate re-asks any clip with the face lost, the head turned or the lips still.

2

Mouth

Phone on a stand at 30–45 cm. Hold the large button, mouth the phrase, release. No sound needed, so ward noise does not matter.

3

Speak

Confident: it speaks in a natural voice the patient chooses, and shows large text. Unsure: it shows the top three to tap.

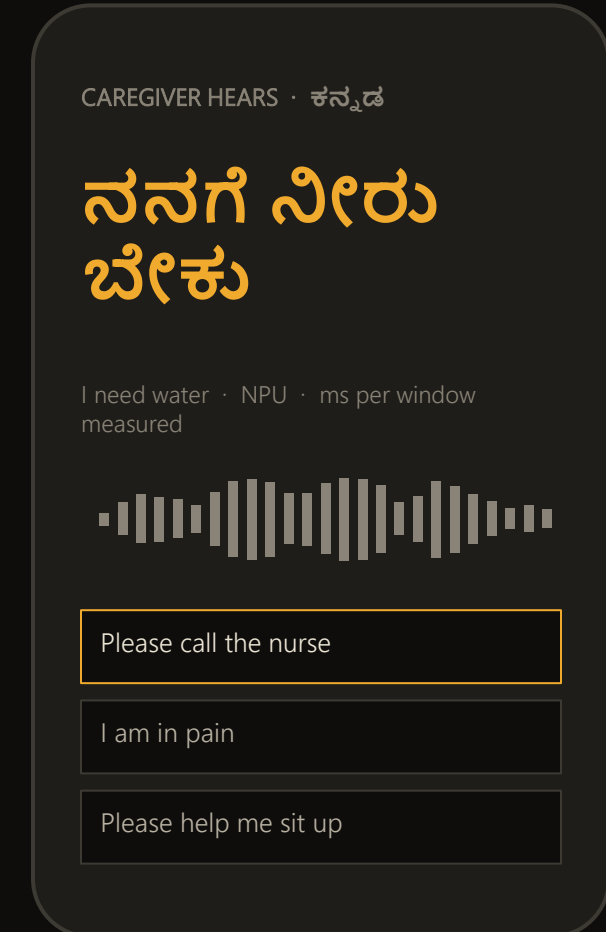
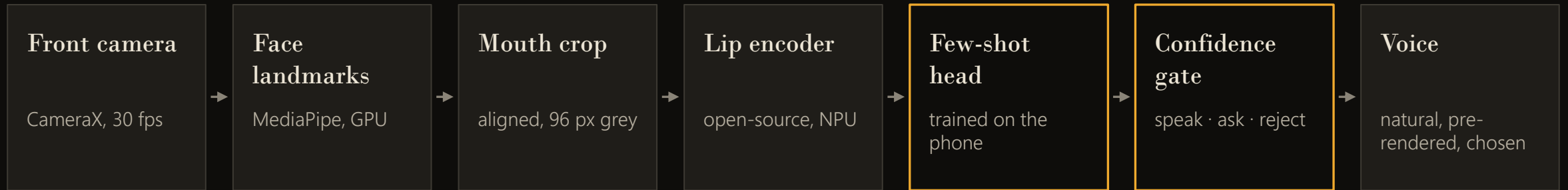


Illustration of the Speak screen

Camera to voice runs entirely on the phone, and the app has no permission to use the internet



Outlined: the part we design and train per person. Video exists only in memory; stored data is a few embedding vectors per phrase.

No INTERNET permission verified in our probe APK: CAMERA is the only permission it requests	18.4 ms on NPU whole encoder for 1.92 s of lips on the iQOO 15's chip (SM8850); target ≤ 700 ms end to end	59.5M params 229 MB fp32 today; int8 on the NPU to fit the < 100 MB target	NPU, GPU, CPU same answer on each path; latency and energy shown side by side
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Built for a ward: hands-free, in the patient's own voice, and it calls the nurse

01

Blink to talk

Two quick blinks start listening, so a patient who cannot press a button can still speak.

02

Their own voice

Before surgery the patient says each phrase aloud once; afterwards Mouna speaks with that recording.

03

Calls the nurse

Each phrase also appears on the caregiver's phone; urgent ones ring it. Direct over Wi-Fi, no server.

04

Learns from mistakes

"Wrong?" takes one tap: the right phrase is spoken and becomes a new sample. A meter shows it improving.

05

Natural voices

4 voices in Kannada, Tamil, Hindi and English, rendered once and played offline.

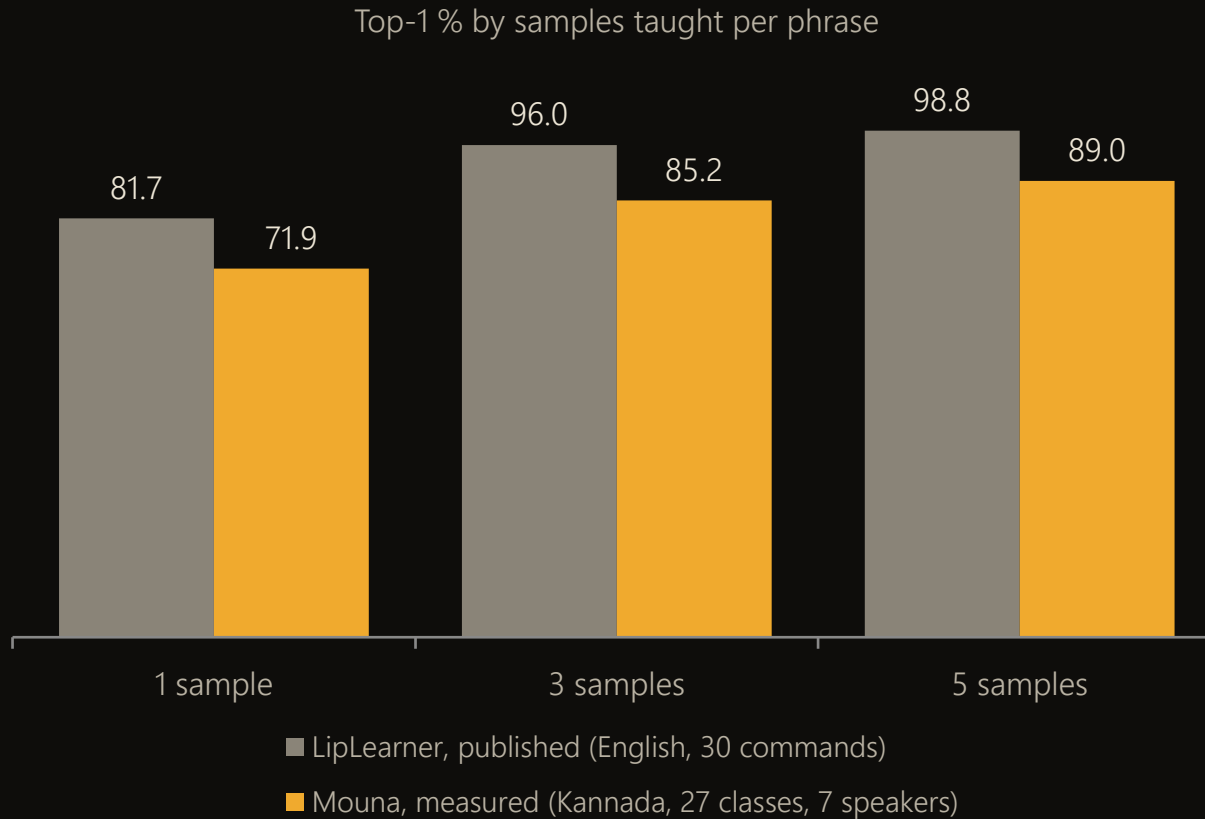
06

Context, safely

After "pain", "medicine" is offered first among the choices. Context reorders; it never speaks on a guess.

All six run in the prototype today (Mouna Lab). Voices: Sarvam Bulbul v3 and AI4Bharat Indic Parler-TTS (Apache-2.0).

Personalised lip reading carries over to Kannada: 89.0% top-1 with five samples per phrase



97.3%

two-word phrases, three samples
(n 327)

94.3%

right when it chose to speak;
otherwise it showed choices

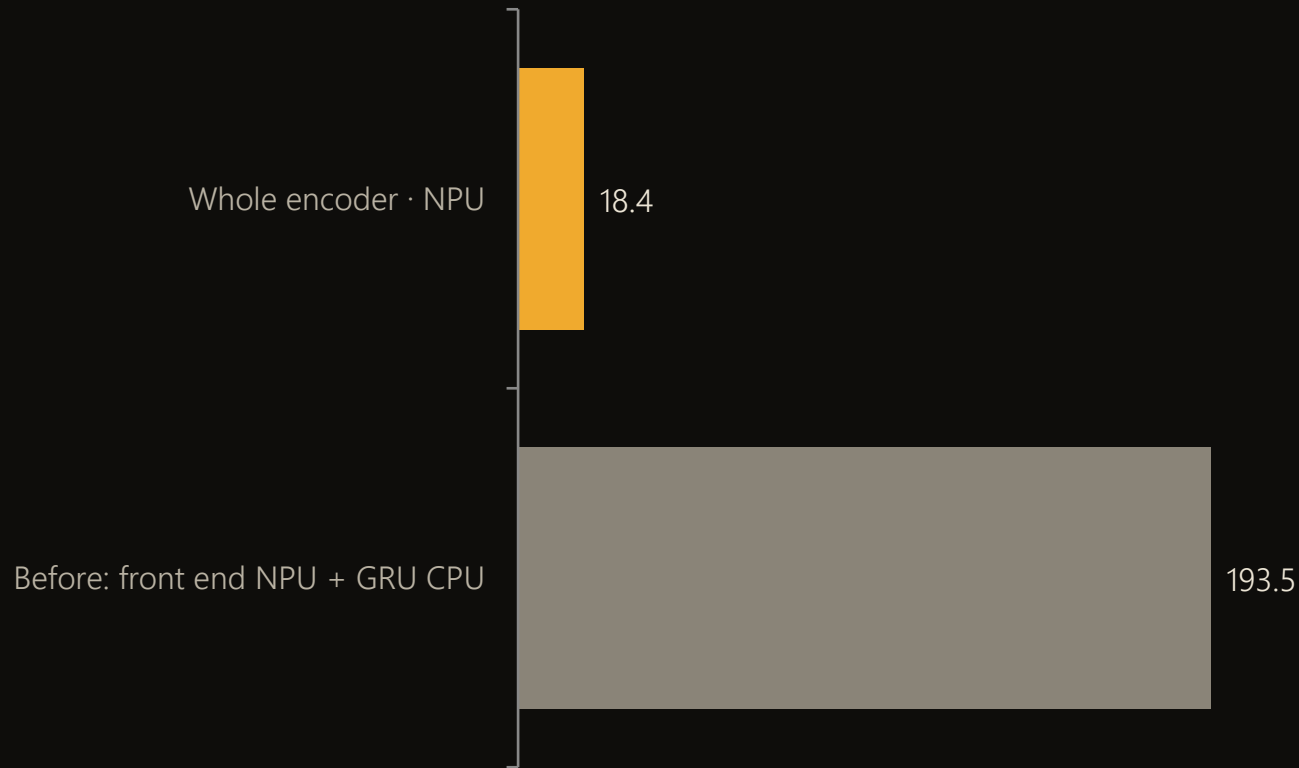
97.2%

right answer among the three
choices, five samples

Single words are the hard case (bare "water" is the hardest), so Mouna's phrases are whole sentences. Next: our own silent recordings, same protocol, evening and next morning.

The whole encoder runs on the Hexagon NPU: 18.4 ms for 1.92 s of lips

Milliseconds per window, Galaxy S26 (SM8850, as in the iQOO 15)



- All 5,265 layers run on the NPU; nothing falls back to the CPU.
- To get there we rewrote the encoder exactly: its 3-D convolution as a 2-D one and its GRU as plain matrix multiplies (difference below 0.000001).
- A fixed 48-frame window made it NPU-ready and more accurate: 86.2% top-1 on Kannada.
- int8 weights: 86.3 MB instead of 230.1 MB, with the same accuracy.

For words it was never taught, Mouna is measured, made careful, and can still be reached by yes and no

23.2% → 7.2%

untaught words spoken as a taught phrase: default vs "Careful" mode. The rest go to choices or "not taught"; Careful still speaks taught phrases right 98.6% of the time.

69.2% → 89.2%

two taught words mouthed one after another ("water · want"): right block by block, vs read against a list of human-written sentences.

2.8%

word pairs on no list that were spoken as a listed sentence. Off-list pairs are offered as the plain words, never as an invented sentence.

Kannada dataset, encoder, 8 of 27 classes taught at random (20 draws per speaker); every other clip is an untaught word. Same teach / test split as the evidence slide.

ASK MODE • WHEN NOTHING TAUGHT FITS

ವಲಿ?

Pain? • nod

ತಲೆ?

Head? • shake

ತೊண்டೆ ಅಲ್ಲತು
ಕழுತ್ತು?

Throat or neck?
• blink twice

ನನಗೆ ಗಂಟಲಾ ಅಥವಾ ಕುತ್ತಿಗೆ
ನೋವಾಗುತ್ತಿದೆ

heard by the nurse in Kannada, and shown live on their phone

30 answers in 8 groups drawn from ICU studies (pain by site, breathing and suction, thirst, comfort, people, fears, questions). Yes: nod, two blinks or tap. No: shake, tap, or wait. One signal is enough.

We build on LipLearner and credit it; what is new is the setting: Indian languages, wards, Android, offline

	On device	Learns your lips	Indian languages	Android	Speaks aloud	Ward safety rules
Mouna	●	●	●	●	●	●
LipLearner (CHI 2023)	●	●	—	—	—	—
Liperty (2026)	●	in progress	—	●	●	—
SRAVI (Liopa)	—	—	—	●	●	clinical
Voice Back (AAC)	●	voice clone	Hindi	●	from taps	—

The wedge: mouth in one language, be heard in another, from human-checked phrase packs, on a phone with no network, with rules that never let it guess in a ward.

On stage a judge teaches Mouna live, and we show the failure case on purpose: it asks instead of guessing

- 0:00 A quiet bed, a whiteboard
- 0:20 Judge teaches three phrases, about 25 s
- 1:10 Mouths in English or Tamil; phone speaks Kannada
- 1:50 Airplane mode, permission list, live telemetry
- 2:20 Untaught: "not sure", then Ask mode finds it
- 2:50 Credit LipLearner; close

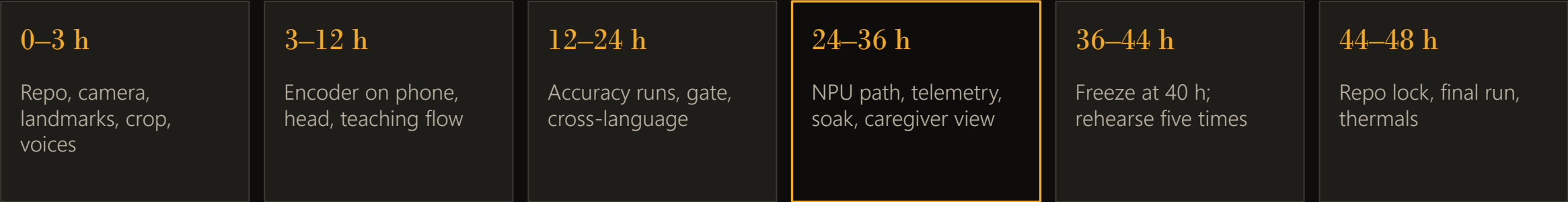
SAFETY RULES

- Speaks only above a per-person threshold and a clear lead over the runner-up.
- Otherwise: top three as large cards, or "not sure, please repeat".
- Urgent phrases (pain, breathing) need a confirm, then alarm and flash.

FALLBACKS

Pre-taught profile · push-to-talk · GPU path if the NPU context fails · 60 s recorded run · second phone with the same build

We know what breaks on a stage, and the 48-hour plan is built around it



Roles: Maadhav, pitch and mouth pipeline · Sachin, app, phrase packs, voices · Nakul, NPU conversion, benchmarks, thermals. NPU cut line at hour 30 (outlined).

Risk	Mitigation
Accuracy below the paper	Spike gate before we commit; five samples; top-3 fallback; geometry plan B
Ops fall off the NPU (Conv3D)	GPU path ready; swap to 2D + temporal conv; report honestly
NPU context error 1007	Pre-warm, cached context binary, retry with backoff, GPU switch on screen
False triggers: smile, yawn	Push-to-talk by default; idle class; measured false speaks per minute
Venue lighting	Own front light; tested under overhead light and backlight

Next: a pilot with a speech-language pathology team, with the limits stated up front

Next

- Pilot with an ENT or speech-language pathology team
- Open-source, human-checked phrase packs in more Indian languages
- Clinical validation before any claim
- Later: “lip banking” before surgery

Limits we state

- Accuracy on real patients is unproven
- Masks, beards, bandages and facial droop block the lips
- A phrase list, not open conversation
- A communication aid, not a medical device

Privacy by design

- No INTERNET permission at all
- Video only in memory; templates encrypted
- Consent screen and one-tap delete
- No real patient in this pitch

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Mouna at a glance, mapped to what a judge looks for

Problem and impact	People who can mouth words but make no sound; ~70,000 new larynx and hypopharynx cancers a year in India; speech disturbance in 51.2% of stroke onsets.
Innovation	Personalised silent-speech phrases with cross-language output (mouth Tamil, heard in Kannada) from human-checked packs; ward safety rules.
Technical depth	MediaPipe landmarks, aligned mouth crop, open-source lip encoder on the Snapdragon NPU, few-shot head trained on the phone, calibrated confidence gate.
Evidence and honesty	Measured on 7 Kannada speakers: 86.2% top-1, 97.3% on phrases, 94.3% right when it speaks; pass lines set before measuring; limits stated.
Feasibility	Built in 21 hours (git history, 3–4 Oct 2026): working prototype live at mouna-spike.vercel.app (Mouna Lab) and tested harness; 48-hour plan with cut lines; GPU fallback; lessons from shipping a VLM on the iQOO 15 NPU.
Use of the device	Whole encoder runs on the Hexagon NPU of the iQOO 15’s chip: 18.4 ms per 1.92 s window, every layer on the NPU (Galaxy S26, Qualcomm AI Hub). Front camera, offline voices, airplane-mode demo.
Responsibility	No network permission, no stored video, consent and delete, not a medical device, prior art credited, pre-built work disclosed.

Every number in this deck, with its source

Kind	Fact	Source
Need	~70,000 new laryngeal and hypopharyngeal cancers a year in India	GLOBOCAN 2024 India fact sheet
Need	51.2% of stroke patients have speech disturbance at onset (34,792 cases)	Mathur et al., Indian stroke registries
Status quo	Cloud lip-reading app SRAVI: 21.8% accuracy in UK critical care	HRA summary of the SRAVI study
Prior art	LipLearner: 81.7 / 96.0 / 98.8% top-1 at 1 / 3 / 5 samples, 30 English commands	Su, Fang, Rekimoto, CHI 2023
Measured	Kannada, 7 speakers, 27 classes: 71.9% / 85.2% / 89.0% top-1 at 1 / 3 / 5 samples (n 1333)	Our harness; Divya P, Mendeley Data 2026, CC BY 4.0
Measured	Two-word phrases 97.3%, single words 83.5% (three samples)	Same
Measured	Right 94.3% of the times it chose to speak; top-3 97.2% with five samples	Same
Measured	Lip geometry alone, no model weights: 75.3% top-1 (three samples)	Same
Measured	Untaught words spoken as a taught phrase 23.2%, 7.2% in Careful mode; two-word sentences 69.2% → 89.2% with a sentence list	Same; 8 of 27 classes taught
Measured	Whole encoder on the Hexagon NPU: 18.4 ms per 1.92 s window, 5265 of 5265 layers on the NPU	Qualcomm AI Hub, Galaxy S26 (SM8850)
Measured	int8 encoder 86.3 MB (fp32 230.1 MB), same Kannada top-1 86.2%	Our harness
Measured	Encoder rebuilt from LipLearner’s MIT Core ML release; 59.5M parameters; matches the original to within 0.00002	Our encoder tools
Measured	Prototype APK requests the CAMERA permission only; the Lab page cannot make network requests (CSP)	aapt2 in CI; HTTP headers
Built	Voice pack: 208 clips (4 voices × 4 languages × 13 phrases), 2.7 MB, rendered once and played offline	Sarvam Bulbul v3; AI4Bharat Indic Parler-TTS (Apache-2.0)

We have shipped on-device AI on this phone before, and we disclose everything that existed before the event

TEAM HIGHEST IN THE ROOM

- VeriTransit, iQOO Chennai City Battle: Qwen3-VL-4B on the iQOO 15 NPU via GenieX
- DriftSense, SEMICON India 2026: 1st runner-up, sub-pixel image registration
- Android in Kotlin and Compose: CameraX, Nearby, BLE (MeshSOS)

DISCLOSURE

Pre-existing: the LipLearner method (cited); open-source MediaPipe, the LipLearner encoder (weight terms being confirmed), LiteRT, ONNX Runtime, the Qualcomm AI stack, Sarvam Bulbul and Indic Parler-TTS voices, with attribution; a small feasibility-spike repository, linked. All Finale code is written in the event window, in a new repository. No patient data; volunteers consent.

SOURCES

- LipLearner, Su, Fang, Rekimoto, CHI 2023, arxiv.org/abs/2302.05907; github.com/rkmtlab/LipLearner
- GLOBOCAN 2024 India fact sheet, gco.iarc.who.int
- Mathur et al., hospital-based stroke registries in India, 2020–22
- HRA summary of the SRAVI study (critical care); liopa.ai
- Fassler et al., 2026, lip-reading after laryngectomy, n = 16
- Divya P, Kannada multi-speaker lip-reading dataset, Mendeley Data 2026, doi:10.17632/zbzrbs89pz.1, CC BY 4.0
- Qualcomm AI Hub, profile jobs on Samsung Galaxy S26 (SM8850)
- Voices: Sarvam AI Bulbul v3; AI4Bharat Indic Parler-TTS (Apache-2.0), rendered at build time
- National Cancer Grid head and neck guideline, 2019, ncgindia.org
- Liperty, github.com/HereLiesAz/Liperty; MELDER, CHI 2024; Auto-AVSR, github.com/mpc001/auto_avsr